

API INTEGRATION & JSON HANDLING SYSTEM

Internship Project Report

Submitted By :-

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Organization

Alfido Tech Internship Program

Technology Stack

Python | REST APIs | JSON | CSV | Requests Library

Submission Date

June 2026

EXECUTIVE SUMMARY

This project demonstrates the implementation of API Integration and JSON Handling using Python. The application communicates with external APIs to collect real-time weather and news information, processes JSON responses, applies filtering logic, and exports structured data into JSON and CSV formats.

The project was designed following modular programming principles, where individual components are responsible for API communication, data processing, file handling, and utility functions. Exception handling mechanisms were implemented to ensure application stability during API failures or unexpected responses.

The final solution successfully fulfills all internship requirements while showcasing practical software development skills used in real-world applications.

PROJECT OBJECTIVE

The primary objective of this project is to understand how Python applications interact with external APIs and process JSON data efficiently.

Goals

- Fetch data using Python Requests Library
- Parse JSON responses
- Apply filtering and search operations
- Implement exception handling
- Store data in JSON format
- Export processed data into CSV format
- Develop a modular and maintainable code structure

PROJECT OVERVIEW

The application integrates with external APIs to retrieve:

Weather Information

The system fetches weather data for multiple cities and stores the results in structured JSON and CSV formats.

News Information

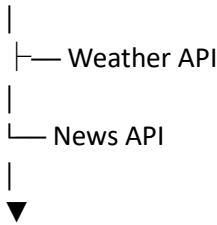
The system retrieves news articles related to Python programming and saves them for future analysis.

Data Processing

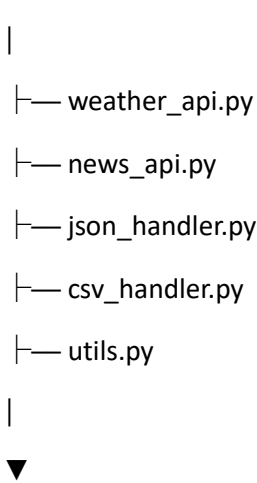
The application filters and processes retrieved data before exporting it into structured files.

SYSTEM ARCHITECTURE

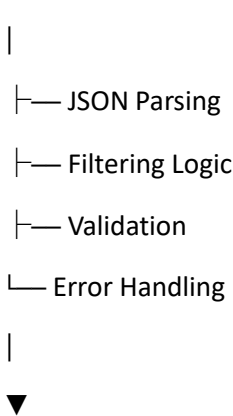
External APIs



Python Application



Data Processing Layer



Output Generation

```
|
|
|— weather_data.json
|— weather_data.csv
|— news_data.json
|— news_data.csv
```

PROJECT STRUCTURE

Task2_API_Integration

```
|
|
|— main.py
|— weather_api.py
|— news_api.py
|— json_handler.py
|— csv_handler.py
|— utils.py
|
|— output
|
| |— weather_data.json
| |— weather_data.csv
| |— news_data.json
| |— news_data.csv
|
|
|— README.md
```

MODULE DESCRIPTION

main.py

Acts as the entry point of the application and coordinates the execution flow.

Responsibilities:

- Program initialization
- API execution
- Function coordination
- Output generation

weather_api.py

Responsible for fetching weather data from external APIs.

Features:

- API requests
 - Weather retrieval
 - Response validation
 - Exception handling
-

news_api.py

Handles news article retrieval.

Features:

- News API communication
 - JSON processing
 - Article extraction
 - Search functionality
-

json_handler.py

Manages JSON operations.

Features:

- Save data into JSON files
 - Read JSON data
 - Data formatting
-

csv_handler.py

Responsible for CSV file generation.

Features:

- CSV creation
 - Data export
 - Structured storage
-

utils.py

Contains reusable helper functions used throughout the application.

IMPLEMENTATION DETAILS

Weather Data Processing

The system retrieves weather information for multiple cities including:

- Delhi
- London
- New York
- Tokyo
- Sydney

The retrieved information includes:

- City Name
- Country
- Temperature
- Wind Speed
- Weather Code
- Day/Night Status

The application then filters cities whose temperature exceeds 25°C.

News Data Processing

The system retrieves Python-related news articles.

Information extracted:

- Article Title
- Source
- Publication Date
- URL
- Description

A keyword-based filtering mechanism identifies articles containing specific keywords such as "AI".

ERROR HANDLING IMPLEMENTATION

The application implements exception handling using Python try-except blocks.

Handled Scenarios:

- Network failures
- Invalid API responses
- Missing JSON fields

- File writing errors

Benefits:

- Improved reliability
 - Better user experience
 - Prevention of unexpected crashes
-

OUTPUT FILES GENERATED**Weather Data JSON**

Contains structured weather information in JSON format.

Weather Data CSV

Contains tabular weather information suitable for spreadsheets.

News Data JSON

Stores news articles in machine-readable format.

News Data CSV

Stores article information in tabular format.

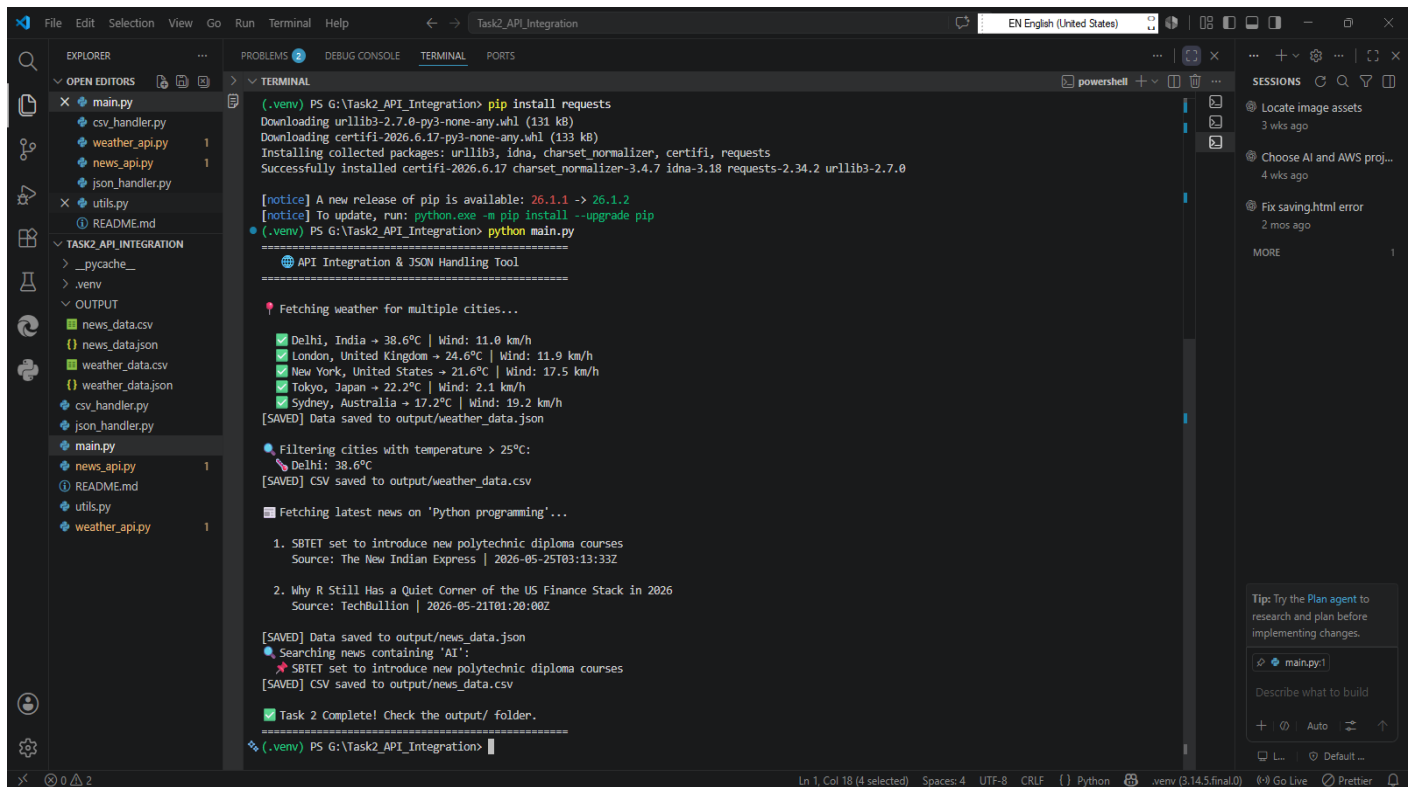
RESULTS

The application successfully:

- ✓ Retrieved live weather information
- ✓ Retrieved news articles
- ✓ Parsed JSON responses
- ✓ Applied filtering logic
- ✓ Exported data to JSON files
- ✓ Exported data to CSV files
- ✓ Handled exceptions correctly
- ✓ Generated structured outputs

SCREENSHOTS

Figure 1: Successful Program Execution



```
(.venv) PS G:\Task2_API_Integration> pip install requests
Downloading urllib3-2.7.0-py3-none-any.whl (131 kB)
Downloading certifi-2026.6.17-py3-none-any.whl (133 kB)
Installing collected packages: urllib3, idna, charset normalizer, certifi, requests
Successfully installed certifi-2026.6.17 charset normalizer-3.4.7 idna-3.18 requests-2.34.2 urllib3-2.7.0

[notice] A new release of pip is available: 26.1.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS G:\Task2_API_Integration> python main.py
=====
API Integration & JSON Handling Tool
=====

🔴 Fetching weather for multiple cities...

✅ Delhi, India → 38.6°C | Wind: 11.0 km/h
✅ London, United Kingdom → 24.6°C | Wind: 11.9 km/h
✅ New York, United States → 21.6°C | Wind: 17.5 km/h
✅ Tokyo, Japan → 22.2°C | Wind: 2.1 km/h
✅ Sydney, Australia → 17.2°C | Wind: 19.2 km/h
[SAVED] Data saved to output/weather_data.json

🔵 Filtering cities with temperature > 25°C:
🔴 Delhi: 38.6°C
[SAVED] CSV saved to output/weather_data.csv

📄 Fetching latest news on 'Python programming'...

1. SBTET set to introduce new polytechnic diploma courses
Source: The New Indian Express | 2026-05-25T03:13:33Z

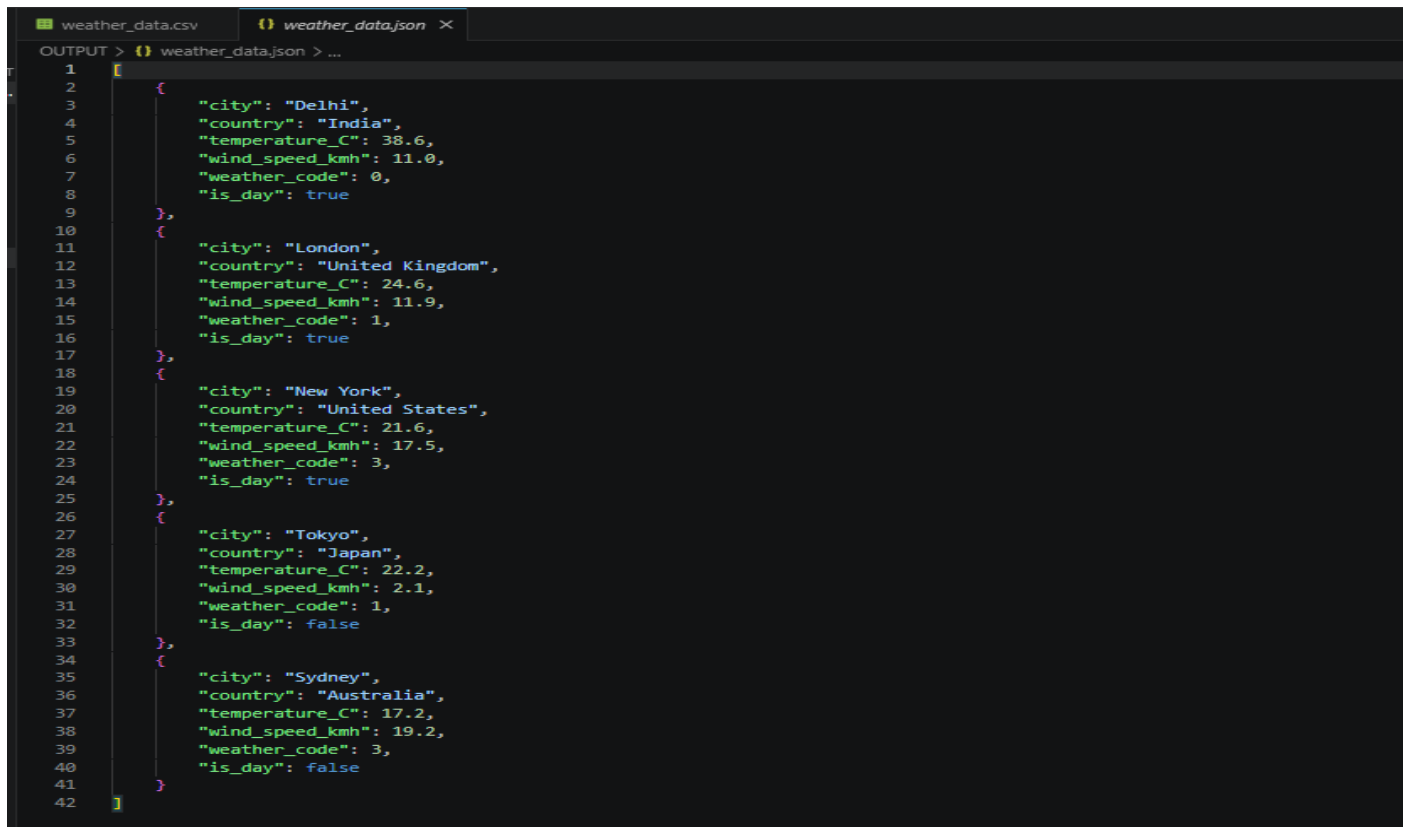
2. Why R Still Has a Quiet Corner of the US Finance Stack in 2026
Source: TechBullion | 2026-05-21T01:20:00Z

[SAVED] Data saved to output/news_data.json
🔵 Searching news containing 'AI':
🔴 SBTET set to introduce new polytechnic diploma courses
[SAVED] CSV saved to output/news_data.csv

✅ Task 2 Complete! Check the output/ folder.
=====
(.venv) PS G:\Task2_API_Integration>
```

This screenshot shows the successful execution of the API Integration & JSON Handling application. The system fetched real-time weather information for multiple cities and retrieved news articles related to Python programming. The retrieved data was processed, filtered according to predefined conditions, and successfully stored in both JSON and CSV formats. The terminal output confirms that all modules executed without errors.

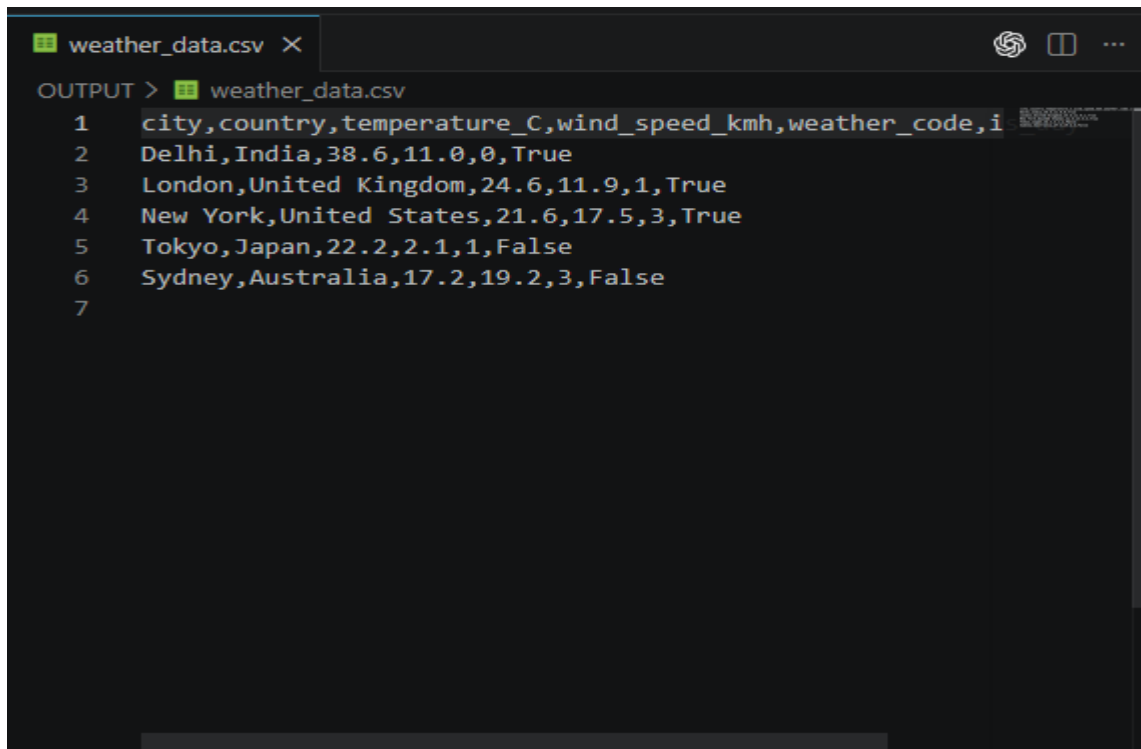
Figure 2: Weather Data JSON Output.



```
1  [
2
3      {
4          "city": "Delhi",
5          "country": "India",
6          "temperature_C": 38.6,
7          "wind_speed_kmh": 11.0,
8          "weather_code": 0,
9          "is_day": true
10     },
11     {
12         "city": "London",
13         "country": "United Kingdom",
14         "temperature_C": 24.6,
15         "wind_speed_kmh": 11.9,
16         "weather_code": 1,
17         "is_day": true
18     },
19     {
20         "city": "New York",
21         "country": "United States",
22         "temperature_C": 21.6,
23         "wind_speed_kmh": 17.5,
24         "weather_code": 3,
25         "is_day": true
26     },
27     {
28         "city": "Tokyo",
29         "country": "Japan",
30         "temperature_C": 22.2,
31         "wind_speed_kmh": 2.1,
32         "weather_code": 1,
33         "is_day": false
34     },
35     {
36         "city": "Sydney",
37         "country": "Australia",
38         "temperature_C": 17.2,
39         "wind_speed_kmh": 19.2,
40         "weather_code": 3,
41         "is_day": false
42     }
43 ]
```

This screenshot displays the generated `weather_data.json` file. The data is stored in structured JSON format, making it easy to process programmatically. Each record contains information such as city name, country, temperature, wind speed, weather code, and day/night status. This demonstrates successful JSON parsing and data storage.

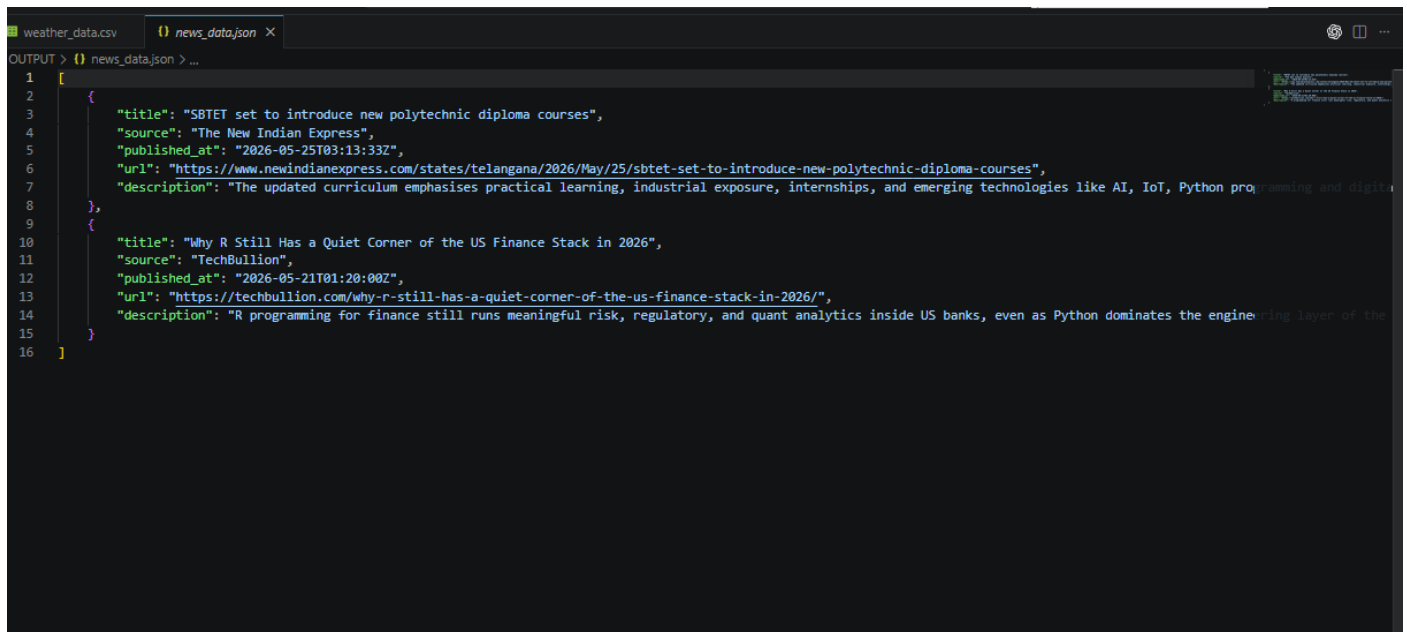
Figure 3: Weather Data CSV Output



```
weather_data.csv X
OUTPUT > weather_data.csv
1 city,country,temperature_C,wind_speed_kmh,weather_code,i
2 Delhi,India,38.6,11.0,0,True
3 London,United Kingdom,24.6,11.9,1,True
4 New York,United States,21.6,17.5,3,True
5 Tokyo,Japan,22.2,2.1,1,False
6 Sydney,Australia,17.2,19.2,3,False
7
```

This screenshot shows the exported `weather_data.csv` file. The weather information retrieved from the API has been converted into tabular format, enabling easy analysis using spreadsheet applications such as Microsoft Excel. This output verifies the successful implementation of CSV file handling and data export functionality.

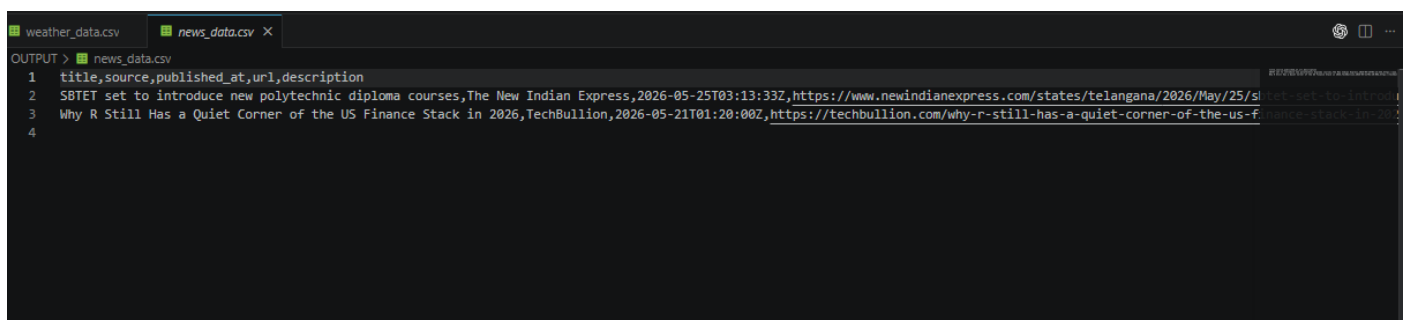
4: News Data JSON Output

A screenshot of a code editor with two tabs: 'weather_data.csv' and 'news_data.json'. The 'news_data.json' tab is active, showing a JSON array of two news article objects. The first object has a title about SBTET diploma courses, and the second object has a title about the US Finance Stack. The code is syntax-highlighted in a dark theme.

```
1 [
2   {
3     "title": "SBTET set to introduce new polytechnic diploma courses",
4     "source": "The New Indian Express",
5     "published_at": "2026-05-25T03:13:33Z",
6     "url": "https://www.newindianexpress.com/states/tehangana/2026/May/25/sbtet-set-to-introduce-new-polytechnic-diploma-courses",
7     "description": "The updated curriculum emphasises practical learning, industrial exposure, internships, and emerging technologies like AI, IoT, Python programming and digital marketing."
8   },
9   {
10    "title": "Why R Still Has a Quiet Corner of the US Finance Stack in 2026",
11    "source": "TechBullion",
12    "published_at": "2026-05-21T01:20:00Z",
13    "url": "https://techbullion.com/why-r-still-has-a-quiet-corner-of-the-us-finance-stack-in-2026/",
14    "description": "R programming for finance still runs meaningful risk, regulatory, and quant analytics inside US banks, even as Python dominates the engineering layer of the industry."
15  }
16 ]
```

This screenshot presents the `news_data.json` file containing news articles related to Python programming. Each article record includes the title, source, publication date, URL, and description. The output demonstrates successful API communication, JSON response processing, and structured data storage.

Figure 5: News Data CSV Output

A screenshot of a code editor with two tabs: 'weather_data.csv' and 'news_data.csv'. The 'news_data.csv' tab is active, showing a CSV file with two rows of news article data. The first row is a header with column names, and the second row contains the data for the first article. The code is syntax-highlighted in a dark theme.

```
1 title,source,published_at,url,description
2 SBTET set to introduce new polytechnic diploma courses,The New Indian Express,2026-05-25T03:13:33Z,https://www.newindianexpress.com/states/tehangana/2026/May/25/sbtet-set-to-introduce-new-polytechnic-diploma-courses,The updated curriculum emphasises practical learning, industrial exposure, internships, and emerging technologies like AI, IoT, Python programming and digital marketing.
3 Why R Still Has a Quiet Corner of the US Finance Stack in 2026,TechBullion,2026-05-21T01:20:00Z,https://techbullion.com/why-r-still-has-a-quiet-corner-of-the-us-finance-stack-in-2026/,R programming for finance still runs meaningful risk, regulatory, and quant analytics inside US banks, even as Python dominates the engineering layer of the industry.
```

This screenshot displays the generated `news_data.csv` file. The news information obtained from the API has been exported into CSV format for easier reading, sharing, and further analysis. The successful creation of this file confirms the effectiveness of the CSV export module.

CHALLENGES FACED

Challenge	Solution
API communication failures	Implemented try-except handling
JSON parsing complexity	Structured data extraction methods
Data storage management	Created reusable JSON and CSV handlers
Code maintainability	Adopted modular programming approach

LEARNING OUTCOMES

During this project, the following skills were developed:

- Python Programming
- API Integration
- REST API Communication
- JSON Parsing
- CSV File Handling
- Error Handling
- Modular Programming
- Software Documentation
- Data Processing
- Problem Solving

FUTURE ENHANCEMENTS

Possible future improvements include:

- Graphical User Interface (GUI)
- Database Integration
- Data Visualization Dashboard
- Scheduled Data Collection
- Real-Time Monitoring System
- Cloud Deployment
- User Authentication

CONCLUSION

The API Integration & JSON Handling System successfully demonstrates the practical use of Python for communicating with external services and processing real-world data.

The project fulfills all internship requirements by implementing API communication, JSON handling, filtering operations, CSV export functionality, and exception handling. Additionally, the modular architecture improves maintainability and scalability, making the project suitable for future enhancements and real-world applications.

This project strengthened my understanding of API-based development, data processing techniques, and professional software engineering practices.